Hierarchical Semantic Invariants Across Transformer Depth

Abstraction, Residual Dynamics, and Layerwise Representation

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Abstract

We implement a controlled framework for measuring hierarchical semantic invariants across Transformer depth. Paper 0.6 reuses Paper 0.5's exact self-attention (SA), feed-forward (FFN), residual, intervention, entropy, artifact, and checkpoint infrastructure. Balanced noun, action, and relation families contain natural-label and arbitrary-label synthetic trees, parent and root queries, three templates, and permuted-parent controls. Crucially, semantic interpretation is gated by held-out completion competence. None of four trained runs passes the preregistered 80% gate: final accuracies range from 1.4% to 15.3%. We therefore report geometry, entropy trajectories, sibling/cross-category replacements, and component ablations as diagnostics of an underfit system, not evidence of learned categories. Same-parent sibling outputs are only modestly closer than cross-category outputs (0.660 versus 0.746 JS bits), and sibling replacement is only slightly less disruptive. Entropy rises from pre-SA to post-block for both parent and root requests. These are useful failure measurements: attractive geometry (neighbor recovery 0.477 versus 0.254 under permutation) is insufficient when the model cannot perform the held-out task. The principal result is consequently methodological and negative—competence must precede semantic mechanism claims.

1 Motivation

Paper 0.5 asks how familiar sequential regularities are represented and transformed across attention, FFN, and the residual stream. N-grams provide a clean statistical probe. Paper 0.6 supplies the semantic counterpart: do hidden states organize concepts as nested invariances, and how does that organization evolve across depth? The goal is not another demonstration that categories can be decoded by a probe, but a mechanistic account of emergence, persistence, transformation, and causal relevance.

2 Relation to Paper 0.5

The papers should share instrumentation but remain separate. Paper 0.5 controls recurrent token/sequential patterns and studies continuation relations. Paper 0.6 controls semantic hierarchy/specificity and studies category and relational invariants. N-gram frequency and next-token predictability become mandatory confounds in Paper 0.6: semantic effects should survive matching, stratification, or residualization against surface statistics.

3 Abstraction as Nested Invariance

Let $x \sim_k y$ mean that x and y are equivalent at abstraction level k. Falcon and penguin are equivalent under bird; falcon and dog become equivalent under animal. This induces nested classes $C_0(x) \subset C_1(x) \subset C_2(x) \dots$. If T_k is a family of substitutions preserving category k, an invariant representation satisfies $r_k(Tx) \approx r_k(x)$ for $T \in T_k$. Higher abstraction is therefore operationally stability under a larger transformation family.

4 Research Questions

At which layers do instance, category, superclass, action, and relation invariants emerge? Are abstraction levels monotonic with depth or do they coexist and re-enter? Does within-category variability contract while between-category separation increases? Does activation geometry correlate with known tree distance? Do residual increments share category-specific directions or subspaces? Which effects arise across attention versus FFN? How much is explained by lexical frequency, syntax, tokenization, or n-gram familiarity?

5 Controlled Datasets

The implemented corpus uses six balanced binary trees: natural-label and arbitrary-label versions of nouns, actions, and relations. Natural examples include falcon/penguin → bird → entity, walk/crawl → movement → act, and above/below → spatial → relation. Synthetic trees use arbitrary labels introduced by definition chains. Each leaf is queried for its parent and root under three template tokens. All labels are single tokens; query/template counts and local training frequency are balanced. A cross-category leaf with the same template supplies activation and motif controls, and shuffled parent identities supply a metric-level negative control. Micro-to-macro coarse-graining remains E9 and is not conflated with the taxonomy experiment.

6 Layerwise Metrics

For layer l, collect hidden state $h_l(x)$ and candidate residual update $\Delta h_l(x) = h_{l+1}(x) - h_l(x)$. For a parent class C, define within-class dispersion D_W, matched between-class dispersion D_B, and normalized separation

$$S_l(C) = \frac{D_B - D_W}{D_B + \epsilon}. \tag{1}$$

Pairwise cosine-distance matrices are compared with exact tree distances by Spearman RSA. Tree recovery is the fraction of leaves whose nearest representational neighbor has the same parent; parent labels are permuted within the matched family/domain for a negative control. Cross-template invariance averages representation cosine for the same item and requested abstraction under three templates. Effective rank and shared-update explained variance measure concentration without assuming that compression is abstraction.

Following the shared metric ontology, we also retain $\|\Delta_l\|$, $\|\Delta_l\|/\|h_l\|$, $\cos(h_l, \Delta_l)$, update–update cosine, and category-conditioned update rank. Positive, orthogonal, or negative geometry remains a refinement, complementarity, or interference *candidate*; geometry alone is never called causal interference. For layer masks above the permuted-control criterion, onset, consecutive persistence, disappearance, and re-entry are computed explicitly rather than assuming monotonicity.

6.1 Conditional semantic-variance decomposition

At each fixed depth and tensor location, realizations are crossed over template context, entity within generator category, category, family, requested parent/root level, model, and seed. We separate template nuisance variance within an entity, entity/member variance within a category, total within-category variance, and between-category centroid variation. Their Fisher-style ratio is between-category variation divided by within-category variation. We separately retain residual covariance rank, mean JS divergence to the category-conditioned output centroid, and within-example entropy. Depth is a refinement axis, not a source of independent samples. Because the competence gate fails, generator memberships are metadata for a negative-control decomposition and are not called learned categories.

7 SA versus FFN

Capture pre/post-attention and pre/post-FFN states using Paper 0.5's exact convention,

$$r'_l = r_l + \Delta_l^{\mathrm{SA}}, \tag{2}$$

$$r_{l+1} = r'_l + \Delta_l^{\mathrm{FF}}. \tag{3}$$

Signed target-logit progress is diagnostic. Causal outcomes use intact-minus-intervened target log probability under zero update, matched batch mean, cross-category matched activation replacement, selective head ablation, and complete FFN-layer ablation. Parent and root queries create a contextual-resolution comparison for the same leaves. This design tests, rather than assumes, that attention retrieves relations and FFNs store concepts.

8 Invariant Taxonomy

Distinguish lexical invariance, syntactic invariance, sequential/n-gram invariance, category invariance, hierarchical invariance, relational invariance, abstraction invariance, and task invariance. A useful summary object is I(layer, abstraction level, transformation family), enabling heatmaps and comparisons across models and semantic families.

9 Controls and Causality

Token length, syntactic role, templates, number of occurrences, and local training frequency are balanced by construction. The semantic atlas uses Paper 0.5's frequency, continuation-entropy, top-probability, and content-hash schema, and records the hierarchy and Paper 0.5 manifest hashes. Natural labels are compared with arbitrary synthetic labels; exact tree metrics are compared with shuffled parent identities; and invariance is tested across templates. Bootstrap intervals use model/seed/family cells rather than raw token occurrences. Linear probes are unnecessary for the primary result and are not used as evidence. Subspace projection and cross-template direction transfer remain for pretrained follow-up.

10 Hypotheses

H1 Hierarchical geometry: activation distances reflect semantic tree distance in a subset of layers. H2 Non-monotonic abstraction: abstraction is not a single early-to-late ladder. H3 Residual commonality: members of a semantic class share reliable residual-update components/subspaces. H4 Cross-domain partial universality: some signatures generalize across nouns, actions, and relations. H5 Surface-statistics insufficiency: matched n-gram and lexical controls reduce but do not eliminate hierarchical effects.

11 Local Rules, Symmetry, and Coarse-Graining

The same hierarchy-recovery metrics can be applied to transformer representations and networks trained with local rules on known hierarchical generators, giving a shared quantitative language without claiming architectural identity. Concepts can also be treated as stable under families of meaning-preserving transformations, connecting the experiment to symmetry/invariance. Residual increments permit tests of low-dimensional local generators. Micro-to-macro descriptions provide a separate, operational analogy with coarse-graining.

12 Implemented Experimental Sequence

E1 constructs the versioned semantic atlas and exact tree-distance matrix. E2 records layer/location hierarchy geometry. E3 reuses Paper 0.5's diagnostic and causal SA/FFN decomposition. E4 evaluates semantic attention motifs against cross-category controls. E5 contrasts natural with arbitrary synthetic hierarchies. E6 compares parent versus root contextual resolution. E7 replicates across nouns, actions, and relations. E8 repeats geometry at training steps 0, 40, 100, and 160. The model matrix is identical to Paper 0.5: widths/layers/heads 32/2/2 and 48/3/3, seeds 11 and 23. E9 micro/macro coarse-graining is deferred.

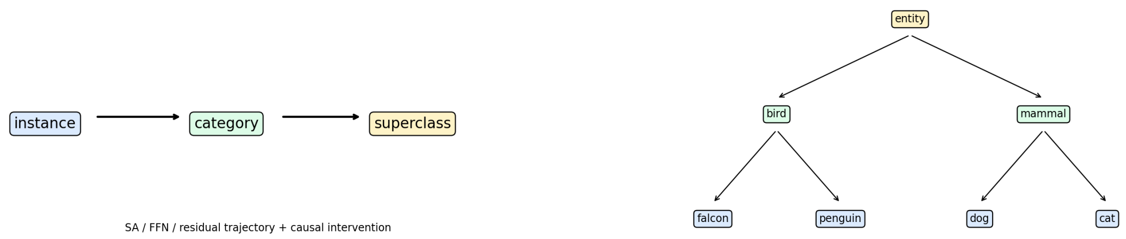

Figure 1: Operational design. Left: abstraction is traced through SA, FFN, and residual updates. Right: a balanced hierarchy with exact distances and substitution classes.

13 Controlled Pilot Results

The artifact set contains 4,320 representation rows, 2,880 causal component rows, 4,320 entropy rows, 5,760 semantic replacement rows, 2,304 checkpoint competence rows, 108 aggregate geometry rows, 13 regenerated figures, and four dense checkpoints for the first model/seed. Final language-model loss ranges from 0.457 to 0.530.

Hierarchy	Separation	RSA	Neighbor	Permuted	Template inv.
Natural	0.142	0.236	0.503	0.233	0.955
Synthetic	0.078	0.152	0.451	0.274	0.964

Table 1: Post-block hierarchy geometry averaged across layers, families, model settings, and seeds. Neighbor is nearest-neighbor parent recovery; Permuted shuffles parent identities under matched representations.

13.1 Competence gate: failed

The held-out completion gate was fixed at 80% before semantic interpretation. Final accuracies are 4.9%, 1.4%, 4.9%, and 15.3% across the four setting/seed runs. Low aggregate language-model loss is therefore not evidence that the requested hierarchy completion was learned. All analyses below are retained as failure diagnostics and must not be read as category evidence.

Natural-label geometry is stronger than arbitrary-label geometry, and both exceed their permuted-parent controls. Across post-block cells, normalized separation is 0.110, hierarchy RSA is 0.194, and parent-neighbor recovery is 0.477 versus 0.254 after label permutation. Cross-template cosine is 0.960. Because competence failed, these values demonstrate precisely why geometry alone is unsafe: they may reflect token, template, or partially trained structure rather than usable semantic categories.

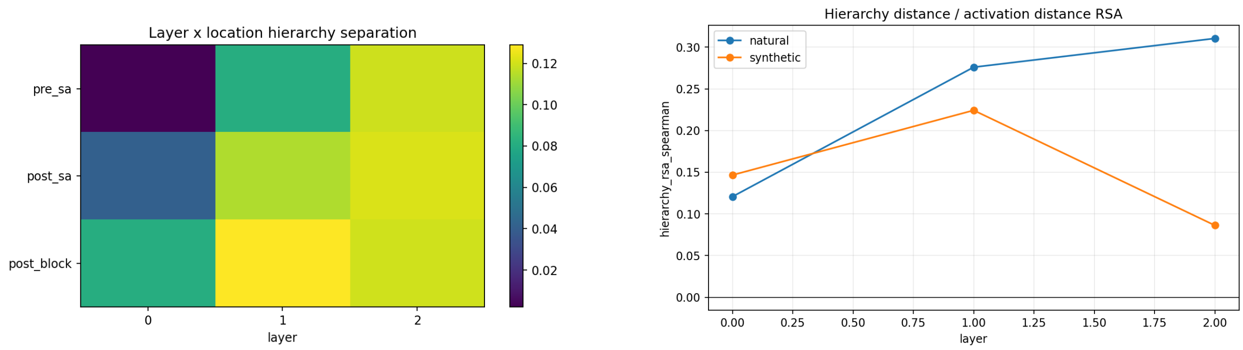

Figure 2: Layerwise hierarchy geometry. Left: normalized separation by tensor location and layer. Right: tree-distance/activation-distance RSA for natural and synthetic labels.

13.2 Equivalence, entropy, and replacement diagnostics

Same-parent sibling output distributions have mean JS divergence 0.660 bits, compared with 0.746 bits across semantic families. This difference is directionally consistent with the designed nesting but is small relative to the distance itself and cannot pass the competence prerequisite. Complete-update replacement shows similarly weak specificity: at layer-0 FFN, sibling versus cross-category output change is 0.241 versus 0.284 JS bits; at layer-0 SA it is 0.205 versus 0.243. Target-probability changes are near zero for sibling donors and somewhat more negative for cross-category donors, but the unlearned target makes this contrast non-semantic.

Decoded uncertainty is non-monotone in the sense required by the shared ontology and, on average, increases through a block. Parent-query entropy is 0.629/0.792/1.107 bits at pre-SA/post-SA/post-block; root-query entropy is 0.596/0.791/1.139. These trajectories reject a universal entropy-reduction story, but they do not reveal abstraction while the behavioral gate is closed. Common residual-component metrics are committed for audit and future competent replication; no common direction is promoted to a semantic invariant here.

13.3 Variance curves are informative controls, not semantic evidence

Across post-block cells, mean JS-to-generator-category centroid decreases from 0.376 to 0.328 to 0.213 bits over layers 0–2, while mean within-example entropy rises from 0.797 to 1.224 to 1.547 bits. The same dissociation seen in Paper 0.5 is therefore measurable here: output distributions can become more consistent across realizations while remaining broad individually. However, the failed completion gate prevents a semantic interpretation.

Total within-category residual variance rises from 28.71 to 43.60 to 57.54, and between-category variation rises from 35.35 to 52.66 to 68.74. Total variance reduction is therefore false. The diagnostic Fisher ratio nevertheless rises from 1.668 to 1.724 to 1.957 because between-category structure grows somewhat faster than within-category dispersion. Template nuisance variance is non-monotone (2.50, 7.62, 5.41), whereas member variance grows (26.20, 35.97, 52.13). These metadata-conditioned curves show why the correct prediction concerns relative task signal, not shrinking total hidden-state variance—and why behavior must arbitrate whether that signal is semantic.

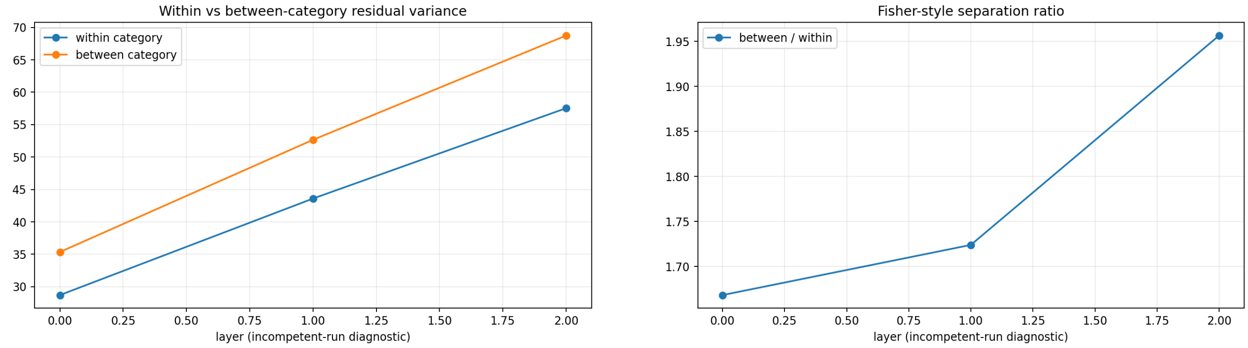

Figure 3: Generator-metadata variance diagnostics at fixed depth. Both within- and between-category residual variation grow; their ratio rises modestly. The competence gate failed, so these are controls rather than semantic evidence.

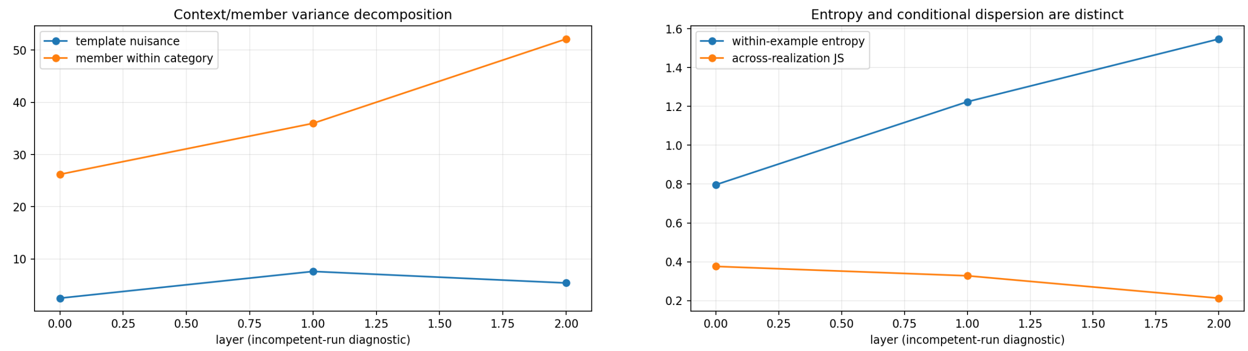

Figure 4: Left: nuisance-template and entity/member components. Right: within-example entropy versus across-realization JS dispersion, retained as distinct quantities.

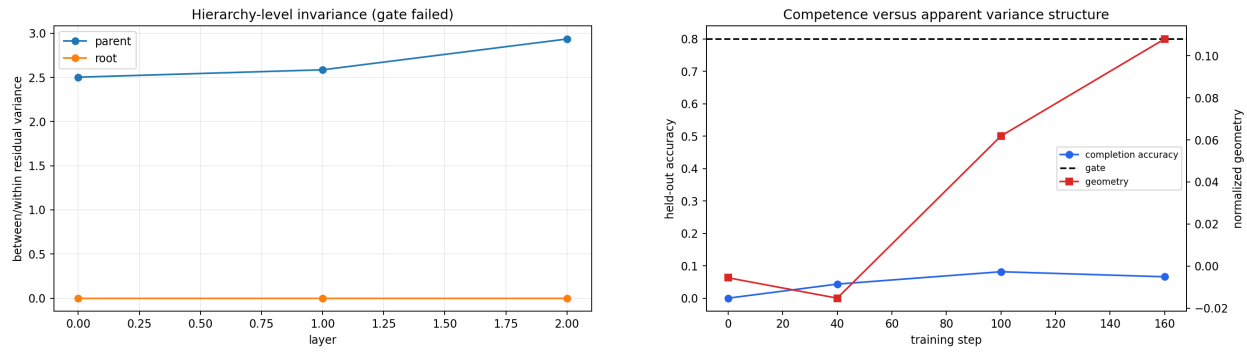

Figure 5: Left: parent/root metadata-conditioned ratios without a monotone abstraction assumption. Right: apparent geometry alongside held-out competence and its frozen 80% gate.

13.4 Dispersion, invariance, and residual commonality

Between-parent distance exceeds within-parent distance on average, while high cross-template cosine shows that the same item/requested level remains stable across the three controlled forms. The invariant criterion is already above the permuted control at layer 0 and persists for all measured layers in both natural and synthetic conditions. This is evidence against a simple late-onset abstraction ladder: part of the geometry is present in the learned embedding/context state before deeper transformation. Shared-update explained variance is reported as residual commonality, not as a literal group generator.

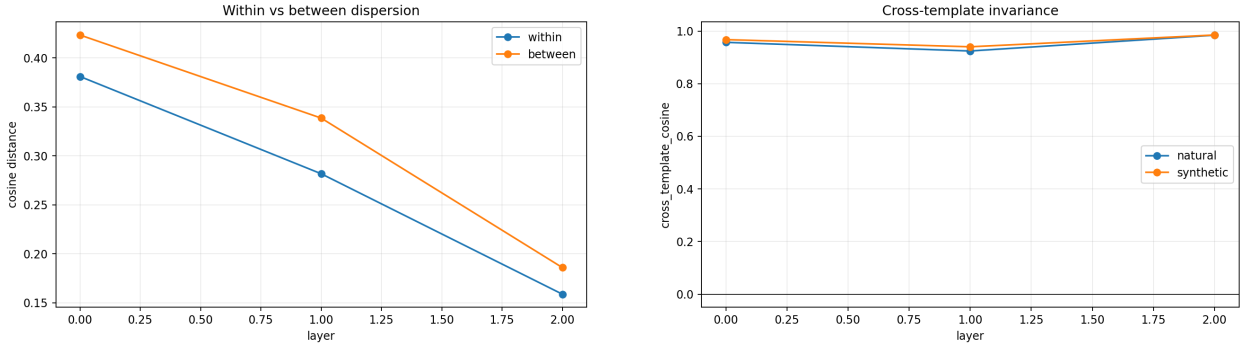

Figure 6: Left: within- versus between-parent cosine dispersion. Right: invariance of an item's representation across matched templates.

13.5 Task-conditioned component responsibility

Zero ablation gives opposite component orderings for the two requested levels. Parent prediction is FFN-heavy (1.117 versus 0.211 nats for SA); root prediction is SA-heavy (0.987 versus 0.671). Matched replacement changes the root ordering (FFN −0.223, SA 0.268), again showing intervention sensitivity. The same leaf therefore recruits different component mixtures under fine versus coarse resolution, supporting task-conditioned coexistence rather than a fixed layer-to-abstraction map.

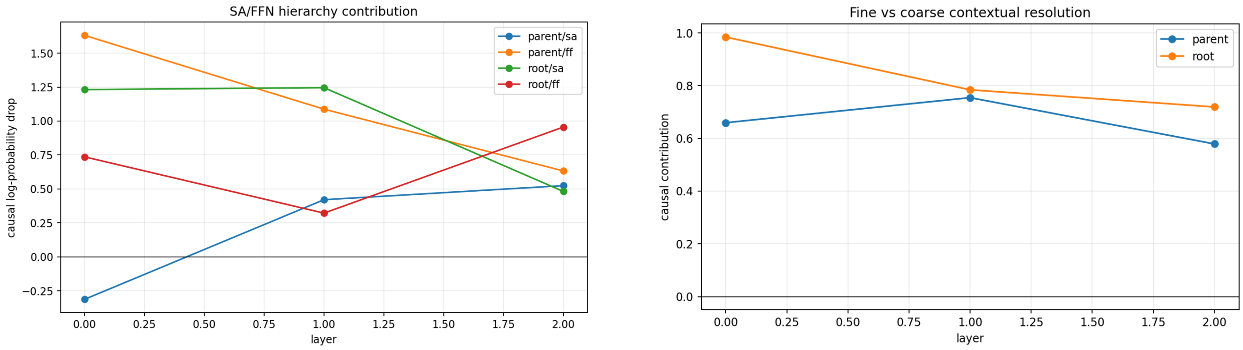

Figure 7: Left: SA/FFN causal contribution by abstraction target. Right: aggregate parent (fine) versus root (coarse) resolution. Signed values are retained.

13.6 Motif null and cross-domain variation

Mean semantic motif specificity is −0.012: within-equivalence attention rows are no more stable than cross-category, position-matched controls. E4 is therefore negative even though hidden-state geometry exceeds permuted control. Across domains, normalized separation/RSA/neighbor recovery are 0.060/0.095/0.427 for actions, 0.120/0.185/0.443 for nouns, and 0.150/0.302/0.563 for relations. H4 is at most partial: the direction generalizes, but magnitude is domain-dependent.

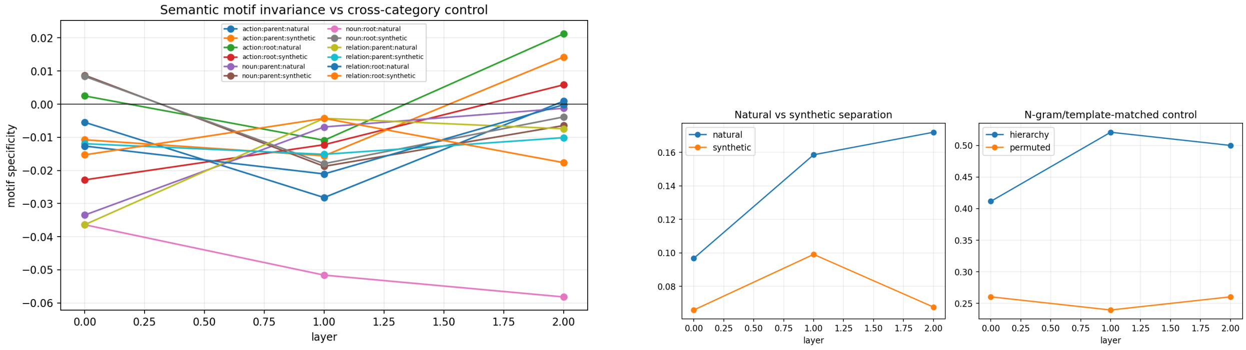

Figure 8: Left: semantic attention-motif specificity, which averages slightly below zero. Right: natural/synthetic geometry and exact hierarchy recovery against the permuted-parent control.

13.7 Development and update organization

Hierarchy separation changes across checkpoints rather than rising as a guaranteed monotone developmental ladder. Category-conditioned residual updates share some variance, but their commonality differs across SA, FFN, and depth. These measurements retain the clocks distinguished in Paper 0: layer depth l is not training time τ.

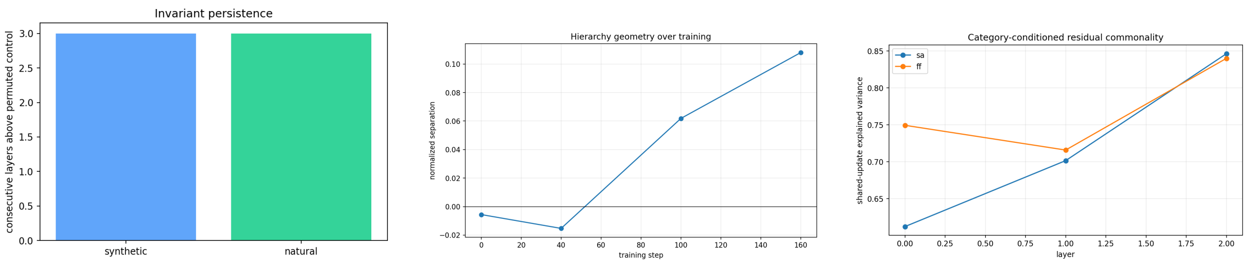

Figure 9: Left: persistence above the permuted-control criterion. Center: hierarchy separation during local training. Right: category-conditioned shared-update explained variance.

14 Series-Level Interpretation

The competence failure changes the series-level conclusion. Representational geometry can exceed a permutation baseline even when held-out behavior is poor; semantic-looking geometry is therefore not sufficient evidence of abstraction. Attention-motif invariance is also absent, while sibling/cross-category replacement contrasts are weak. Paper 0.5's predictive-equivalence result is stronger because its families are behaviorally realized and equivalent replacement preserves an actually learned continuation. Paper 1 may use Paper 0.6 features only as frozen negative controls until a competent replication exists. No semantic feature from this run authorizes persistent learning.

15 Reproducibility and Reuse

The exact command is

```
python -m cl.experiments.paper06_abstraction --steps 160
  --checkpoints 0 40 100 160 --seeds 11,23 --device cpu
```

The manifest records the hierarchy hash, Paper 0.5 manifest hash, resolved model matrix, and artifact hash. Per-run metadata records the source commit/dirty flag, command, model/data identity, seed, device/dtype, package versions, timestamp, and corpus hash. The committed reuse map identifies every shared module and extension. Tests cover tree paths/distances, parent/depth/sibling consistency, deterministic generation, balanced templates, known-geometry recovery, Paper 0.5 join determinism, trace parity, residual identities, intervention locality, and serialization.

16 Falsification, Contribution, and Limitations

The controlled experiment fails its behavioral competence gate and finds no semantic attention-motif invariant. It nevertheless recovers above-permutation geometry, demonstrating that such geometry can be misleading in an underfit system. The supported contribution is a reusable competence-first causal/geometric framework and a documented negative result, not evidence for learned hierarchy structure.

Only two small locally trained architectures and two seeds are studied. All labels are single tokens and templates are controlled tokens rather than natural paraphrases. Balancing removes measured Paper 0.5 confounds but does not estimate their effects in natural language. Activation replacement can be off-manifold; generic layer utility is not eliminated; head 0 is diagnostic rather than corrected search; no subspace projection or direction transfer is attempted. Paper 1's retained pretrained audit concerns graph recovery rather than these semantic hierarchies and therefore does not satisfy the required pretrained replication. Micro-to-macro language, direct natural-corpus residualization, mixed-effects inference, unseen-paraphrase transfer, and cross-template causal transfer remain required.

17 Conclusion

Treating abstraction as invariance under nested substitutions turns "abstraction across layers" into a quantitative causal problem, but behavioral competence is its entry condition. Here that condition fails decisively. Geometry exceeds a permuted control, sibling outputs are only modestly closer than cross-category outputs, causal replacement specificity is weak, and semantic attention motifs are absent. These measurements are diagnostics of an underfit model, not hierarchy evidence. A future run must first pass identity-disjoint completion, then repeat representational and causal tests before any Paper 1 or Paper 0.7 learning claim is enabled.